3 (x86)

1

1

```
-u
0780:0000 B87E07        MOV     AX,077E
0780:0003 8ED8          MOV     DS,AX
0780:0005 33F6          XOR     SI,SI
0780:0007 B90A00        MOV     CX,000A
0780:000A 8A840A00      MOV     AL,[SI+000A]
0780:000E 3C3C          CMP     AL,3C
0780:0010 723A          JB      004C
0780:0012 3C46          CMP     AL,46
0780:0014 7213          JB      0029
0780:0016 3C50          CMP     AL,50
0780:0018 7216          JB      0030
0780:001A 3C5A          CMP     AL,5A
0780:001C 7219          JB      0037
0780:001E 3C64          CMP     AL,64
```

```
0780:0020 721C          JB      003E
0780:0022 3C64          CMP     AL,64
0780:0024 741F          JZ      0045
0780:0026 EB24          JMP     004C
0780:0028 90            NOP
0780:0029 FE061400      INC     BYTE PTR [0014]
0780:002D EB1D          JMP     004C
0780:002F 90            NOP
0780:0030 FE061500      INC     BYTE PTR [0015]
0780:0034 EB16          JMP     004C
0780:0036 90            NOP
0780:0037 FE061600      INC     BYTE PTR [0016]
0780:003B EB0F          JMP     004C
0780:003D 90            NOP
0780:003E FE061700      INC     BYTE PTR [0017]
```

```
0780:0042 EB08          JMP     004C
0780:0044 90            NOP
0780:0045 FE061800      INC     BYTE PTR [0018]
0780:0049 EB01          JMP     004C
0780:004B 90            NOP
0780:004C 46            INC     SI
0780:004D E2BB          LOOP    000A
0780:004F B8004C        MOV     AX,4C00
0780:0052 CD21          INT     21
```

2 ID array S6——S10

```
-d 0 18
077E:0000  32 32 33 36 31 31 34 39-36 38 4C 45 54 5A 49 58   2236114968LETZIX
077E:0010  63 3F 64 50 00 00 00 00-00                        c?dP.....
```

3 ID array S6——S10

```
-d 0 18
077E:0000  32 32 33 36 31 31 34 39-36 38 4C 45 54 5A 49 58   2236114968LETZIX
077E:0010  63 3F 64 50 02 02 03 02-01                        c?dP.....
```

4

; empty asm file

title I love asm

data segment

ID db '2 236 114 968'

```
    array    db     76, 69, 84, 90, 73, 88, 99, 63, 100, 80
    S6       db     0      ; 60~69
    S7       db     0      ; 70~79
    S8       db     0      ; 80~89
    S9       db     0      ; 90~99
    S10      db     0      ; 100
data ends


code segment
    assume cs:code, ds:data
    main     proc
        ; assign the data segment base address to DS
        mov    ax, data
        mov    ds, ax

        ; TODO ...
       ; | add your code between arrows |
       ; v ---------------------------- v
        xor    si, si
       mov    cx, 10
startloop:
        mov    al, [array + si]
        cmp    al, 60
        jb     next
        cmp    al, 70
        jb     add_S6
        cmp    al, 80
        jb     add_S7
        cmp    al, 90
        jb     add_S8
```

```
        cmp     al, 100
        jb      add_S9
        cmp     al, 100
        je      add_S10
        jmp     next
add_S6:
        inc     S6
        jmp     next
add_S7:
        inc     S7
        jmp     next
add_S8:
        inc     S8
        jmp     next
add_S9:
        inc     S9
        jmp     next
add_S10:
        inc     S10
        jmp     next
next:
        inc     si
        loop    startloop
        ; ^  ---------------------------  ^
        ; |              The END                |

        ; method 2: return to dos
        mov     ax, 4c00h
        int     21h
    main    endp
code ends
```

end main

2

1 1 .lst

```
FF BSMicrosoft (R) Macro Assembler Version 5.00                 4/16/25 23:01:55
I love asm                                                      Page     1-1


                        1                       ; empty asm file
                        2
                        3                       title I love asm
                        4
                        5 0000                          data segment
                        6 0000  32 32 33 36 31 31 34        ID   db   '2236114968'
                        7       39 36 38
                        8 000A  31 34 39 36 38                X    db   '14968'          ; 
                                                ??景1：5位，应该汇编5次
                        9                               ;X   db   '2236114968'      ; 
                                                ??景2：10位，应该汇编6次
                       10 = 0005                                len  equ  ($ - X)
                       11 000F                          data ends
                       12
                       13 0000                          code segment
                       14                              assume cs:code, ds:data
                       15 0000                                 main    proc
```

```
                        ; assign the data segment base
address to DS
0000  B8 ---- R               mov   ax, data
0003  8E D8                   mov   ds, ax

                        ; TODO ...
                          ; | add your code betwe
en arrows |
                          ; v ------------------
--------- v
                        if len le 5
                                rept len
                                      add ax,
ax
                                endm
0005  03 C0             1                       add ax,
ax
0007  03 C0             1                       add ax,
ax
0009  03 C0             1                       add ax,
ax
000B  03 C0             1                       add ax,
ax
000D  03 C0             1                       add ax,
ax
                          endif

                        ; ^ --------------------------
- ^
                          ; |         The END
                          |

                        ; method 2: return to dos
000F  B8 4C00                 mov   ax, 4c00h
0012  CD 21                   int   21h
0014                        main    endp
0014                    code ends
0
```

2 1

```
-u
077F:0000 B87E07          MOV     AX,077E
077F:0003 8ED8            MOV     DS,AX
077F:0005 03C0            ADD     AX,AX
077F:0007 03C0            ADD     AX,AX
077F:0009 03C0            ADD     AX,AX
077F:000B 03C0            ADD     AX,AX
077F:000D 03C0            ADD     AX,AX
077F:000F B8004C          MOV     AX,4C00
077F:0012 CD21            INT     21
```

3 1 X

```
-d 0 e
077E:0000  32 32 33 36 31 31 34 39-36 38 31 34 39 36 38      2236114968149 68
```

4 2 .lst

```
                                ; TODO ...
                                  ; | add your code betwe
                        en arrows |
                                  ; v ------------------
                        --------- v
                                  else
                                        rept 6
                                              add ax,
                        ax
                                        endm
28 0005  03 C0                  1                               add ax,
                        ax
29 0007  03 C0                  1                               add ax,
                        ax
30 0009  03 C0                  1                               add ax,
                        ax
31 000B  03 C0                  1                               add ax,
                        ax
32 000D  03 C0                  1                               add ax,
                        ax
33 000F  03 C0                  1                               add ax,
                        ax
                                  endif

                                ; ^ --------------------------
                        - ^
                                  ; |          The END
                                |
```

5 2

```
-u
0780:0000 B87E07        MOV     AX,077E
0780:0003 8ED8          MOV     DS,AX
0780:0005 03C0          ADD     AX,AX
0780:0007 03C0          ADD     AX,AX
0780:0009 03C0          ADD     AX,AX
0780:000B 03C0          ADD     AX,AX
0780:000D 03C0          ADD     AX,AX
0780:000F 03C0          ADD     AX,AX
0780:0011 B8004C        MOV     AX,4C00
0780:0014 CD21          INT     21
```

6 2 X

```
-d 0 13
077E:0000  32 32 33 36 31 31 34 39-36 38 32 32 33 36 31 31   2236114968223611
077E:0010  34 39 36 38                                       4968
```

7

```
; empty asm file

title I love asm

data segment
    ID    db    '2236114968'
    ;X    db    '14968'              ;    1 5        5
    X     db    '2236114968'         ;    2 10       6
    len   equ   ($ - X)
data ends

code segment
    assume cs:code, ds:data
    main    proc
        ; assign the data segment base address to DS
        mov   ax, data
```

```
        mov     ds, ax

        ; TODO ...
        ; | add your code between arrows |
        ; v ---------------------------- v
        if len le 5
            rept len
                add ax, ax
            endm
        else
            rept 6
                add ax, ax
            endm
        endif

        ; ^ ---------------------------- ^
        ; |              The END                  |

        ; method 2: return to dos
        mov     ax, 4c00h
        int     21h
    main    endp
code ends
end main
```